

value chain. And these are not mutually exclusive possibilities; the company could also combine two or more of them. The firm could also secure better terms from suppliers if it could provide some IP coverage to those suppliers in areas where the suppliers are exposed. Similarly, IP rights could become additional items in negotiations with customers and distribution partners to achieve better terms.

The clear message of figure 4-5 is that firms should not only think about their IP situation in their own business but also consider IP risks in other portions of the value chain.

Patent maps are snapshots of a company's situation at a single point in time. These proactive approaches become even more interesting in a dynamic context, which will be explored further in the next section. The question of whether to protect ideas, and how much to protect them, will be shown to vary over the technology life cycle.

THE DYNAMICS OF IP MANAGEMENT: LINKING IP TO THE TECHNOLOGY LIFE CYCLE

One thing that is well known and well accepted among those who study or work in technology is that technology changes rapidly. Some have defined a technology life cycle (TLC) to help describe the underlying patterns in the constantly changing world of technology. What has not been done to date is to link the management of IP and the business model to the TLC.

The proper management of IP should vary with the stage of the relevant technology in the TLC underlying that IP. Currently, most companies use a one-size-fits-all approach to managing IP, effectively ignoring the underlying technology life cycles. This is a mistake. IP management should align with the stage of the technology in question and help shape the subsequent stages of that technology's development. First, we will examine the life cycle concept in some detail and introduce the IP life cycle model. Then we will explore the implications of this model for the differing ways to develop business models and manage the intellectual property associated with the technology.

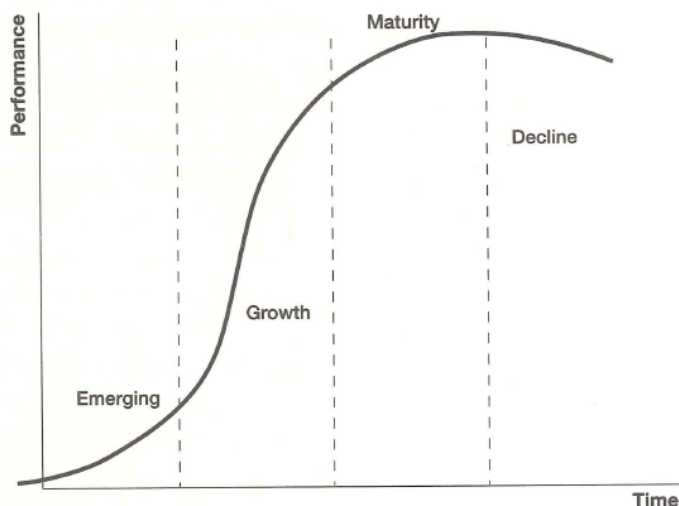
The technology life cycle is one of the most fundamental aspects of managing technology. The idea goes back to the seminal research of William Abernathy of Harvard Business School and James Utterback of MIT, and has been widely embraced by other scholars since.⁷ This line of research has shown that technology does not simply develop at a single, straightforward pace. Instead, there is an initial period where a wide variety of technologies vie for acceptance in the market, a later period where the winning or "dominant design" technology establishes itself in the market, a third period where the technology matures, and a final period in which it becomes obsolete.

The notion of technology life cycles can be well explained with a simple S curve (or "logistic curve," as it is sometimes called). In figure 4-6, the performance of a technology is graphed on the y-axis, while the x-axis shows the time since the technology first appeared. Four distinct stages in the technology's evolution may be observed.

In the first stage, the technology is just beginning to emerge. Its rate of improvement is modest at best, and it takes some amount of time to eke out even this modest performance advance. This stage comes be-

FIGURE 4-6

Stages in the technology life cycle



fore the emergence of a dominant design, so there are a wide variety of potential technologies. In automobiles, for example, there was a time in the very early 1900s when gasoline engines vied with steam and electrical engines to be the dominant power plant for motor vehicles.

The second phase is the growth phase; this is where technologies catch fire and grow explosively in performance. In the terms of academic scholars, the "dominant design" of the technology has taken root, causing the industry to focus its innovative efforts on how to advance the technology within the dominant design. Customers now find the technology to be a powerful solution for at least one pressing problem they face, and companies begin to incorporate the technology into portions of their business. The market then takes off, generating very rapid growth in units and in revenues. To stick with the automotive example, the gasoline engine became the dominant design for the power plant of the car, so innovations began to focus on how to increase the horsepower of the engine. Henry Ford's Model T, and the later Model A, are examples of products in this phase.

The third phase of the curve is the mature phase, where the market growth slows down and the industry reaches its maximum levels of revenue for the technology. At this stage, the technology is well understood, the leading players are well established, and there is little or no entry by new firms into the industry. The technology itself begins to subdivide, creating new niches of opportunity for novel applications of the technology, and penetration is now widespread around the globe. Within the auto industry, this might correspond to Alfred Sloan's segmentation of the auto market into different categories of vehicles and price points. More recently, the rise of sport-utility vehicle and minivan categories could be seen as outcomes of a mature phase of the industry. Customers now show real interest, as the technologies are proven and reliable. Customers themselves are able to innovate by creating new ways of applying the technology to their business problems.

The fourth phase is the declining phase of the technology. In this phase, the technology's rate of performance improvement reaches a plateau. Often, a newer technology has arisen to take the place of the earlier technology, substituting for the functions performed by the earlier technology. While cars continue to grow and advance, albeit

slowly, there are examples of declining technologies within the industry. One such decline was clearly seen in the response of U.S. tire manufacturers to the advent of radial tires coming into the industry from Europe. Relative to the performance offered by radials, bias-ply tires simply couldn't advance much further and eventually conceded the entire market to the newer belt technology.⁸

This view of a technology life cycle is necessarily a simplification of the complexities of technological evolution. But the model shows some important features that will be important for the analysis of crafting business models and managing IP. One important feature is the necessary emergence of a dominant design to spur the growth of the technology in the market. Another important feature is that technological performance is of critical importance early on, while quality, volume, and cost become more critical to success later on. A third important feature is the high number of newly entering firms in the first two stages and the lack of such entries in the final phases.

Perhaps the overarching implication, though, is that companies cannot manage technology in the same way throughout each of the phases of the cycle. The key success factors for the technology in one phase are *not* the key factors in a different phase. Instead of using a one-size-fits-all model of technology management, a company instead must adapt its management of technology to the phase of the cycle in which it competes.

THE IP LIFE CYCLE MODEL

The very same logic of avoiding one-size-fits-all needs to be applied to business models and the management of intellectual property as well. Just as one should not manage technology in the same way in each phase of the technology life cycle, so too one should not manage IP covering that technology the same way throughout its legal life. As with the management of technology, the management of IP should be tailored to the phase of the technology life cycle that the IP covers. I call this tailored approach the "IP life cycle model." Here is how the model works:

- In the initial stages of a new technology, companies must invest in creating IP and choose the best method to protect

that IP. That choice will depend on the role that the new technology might play in the company's business model.

- In the next phase, the company will deploy the technology and go to market. Options to take the technology to market include partnering to obtain distinctive capabilities required to complete the offering and accessing necessary complementary assets to support the offering.
- In the third phase, the company will consider how to harvest the fruits of the technology. While this certainly includes using the technology within the company's own business, the possibilities extend well beyond, to include competitors, customers, suppliers, and third parties in other markets. There are both revenue and profit opportunities that promote external licensing or spin-offs in many cases, as well as strategic considerations that might limit the outside licensing of the technology in other cases.
- In the final phase, the company will manage its exit from the technology. This might be forced by the expiration of legal protection for the IP (note that while patents and copyrights expire, trade secrets do not). Or the exit might also be motivated by the introduction of a new, improved technology that replaces the earlier technology. In these cases, the legal protection for the IP may extend well beyond its value in a use within the original business model. However, the IP may remain quite valuable to another company's business model in a different use.

Let's explore this process in more depth and see how IP might be managed uniquely within each stage.

Emerging Phase

In the initial stages of a new technology, the management of IP will depend greatly on whether the technology fits well with an existing business model. If it doesn't, the business model that makes the best use of the technology may not be apparent and will have to be discovered.

In the former instance, the management of IP is straightforward. The company should seek as much IP protection as it can afford, and it should aggressively develop the technology. When Texas Instruments developed its microprocessor patents, which we saw in the previous chapter, it already had a strong original-equipment-manufacturer (OEM) business model to sell chips to the consumer electronics and computer industries. TI could make these investments with the knowledge that the other elements of its business model already fit well with the technology. Important aspects of the business model—such as the distribution system, the manufacturing and operational processes, and the service and support elements—act as complementary assets in this phase. The brands and trademarks already owned by the firm may extend to cover the new technology. The fit with such complementary assets provides additional assurance that the company will be able to profit from the investment in the technology.⁹

When Apple began the development of its iPod player, it faced many risks. Could its player compete with the other MP3 format players that were already out there? Could its iTunes store, which used a proprietary format to download music, become the preferred place for content owners to list their titles? If Apple could engineer a better user experience, would its more proprietary business model (when compared to MP3 players using free music services that downloaded music illegally) fit with downloadable music?

Apple essentially bet that the answer to these challenges was yes. The company's business model is centered around offering a superior user experience as a value proposition. Apple knew from its many years in the PC business that its Macintosh technology was easier to use and easier to learn, and that Mac users believed they were receiving a superior experience. Apple could bring a similarly superior user experience to downloaded music, played on a handheld device. Moreover, its iPod used many other elements of its business model in operations, distribution, marketing, and sales.¹⁰ Its IP is strongly protected, since Apple can keep the internal elements of its solution proprietary, enabling it to operate with trade secrets, patents, copyrights, and trademark protections. Apple's surrounding complementary assets in its business model provided further protection,

which was one reassuring aspect of its business model to the music companies that would provide the content.¹¹

But there will be other times when a technology does not fit with any established business model and a new business model must be created to commercialize the technology effectively. When Kodak first started working on digital photography, it quickly mastered the technology but struggled to figure out where it could make money. There was no equivalent to the film that Kodak used as its primary source of profits from chemically based photography.

Closer to the iPod space, RealNetworks developed some excellent technology to act as a streaming media player, able to play both MP3-formatted music and digital video content. Its RealAudio and RealPlayer, respectively, enable users to play music and video in a variety of formats. RealNetworks was a start-up and had to figure out a business model that would best allow it to compete with its technologies. The company experimented with a variety of approaches, from selling the standalone player, to bundling it in with subscriptions, to providing specific audio and video content, to creating radio over the Internet, and, later, to enabling computer gaming over the Internet.

In the beginning, Real needed to establish itself as the first and biggest streaming media technology and to try to become the industry standard. Its IP strategy was to brand its player and open up its interfaces while retaining proprietary control over the internal code. But the key strategic goal was to get its RealAudio software established as the *de facto* standard (i.e., the dominant design) for streaming audio content, and later, to get the RealPlayer software similarly ensconced as the *de facto* standard for streaming video content.

In this emerging phase, when a company lacks an appropriate business model at the outset, and there is no dominant design that determines how the technology will unfold, IP protection is secondary to the strategic mission of finding the business model that can best commercialize the technology. Indeed, a company often does not know the best use of a technology at the outset, and so it may be unclear how best to protect that technology for that use. There is little value in very strong protection of a technology that has lost the race to be the dominant design.

Growth Phase

In the next phase, the company will deploy the technology and go to market through a business model. If the company's technology wins the competition to be the dominant design, its technology and business model will likely be widely copied throughout the industry. Winning this competition requires successfully practicing the technology and developing the considerable know-how required to incorporate the technology into the customers' own business, often fitting the technology into a larger system. Significant tacit knowledge is built in doing this.

If the company's technology has not won, there is strong research that shows that it usually is futile to hang on to the approach that lost the race to be the dominant design.¹² Instead, it is best to either withdraw from the industry altogether or shift over to the winning design. In the latter case, the company needs to access the IP of the dominant design, as well as the associated know-how needed to practice the technology effectively. While reverse engineering and hiring away employees from the winning rival firm may help, the losing firm's IP may also be enlisted here. That IP may provide a bargaining chip to gain a license on attractive terms to the winning technology. Alternatively, the losing firm may prefer to exit the industry and receive some licensing revenues for its IP, a sort of consolation prize for losing the race to be the dominant design.

For the winning technology, intellectual property management in this phase takes on a dual character. The first portion of IP management was to support the company in its ability to make its technology become the dominant design. This may have caused the firm to consciously share much of the technology with other firms, in hopes of recruiting more firms to rally around its technology. In this second phase of the IP life cycle, the second portion of IP protection comes into play. Here, the firm seeks to capture a portion of the value created by the now dominant design.

There is a natural tension between these two elements of IP management in this phase.¹³ As has been observed in many technology competitions, one can win the battle of becoming the dominant design and still lose the war to capture the value from this victory.

Rambus, the pure-play DRAM IP company we saw in the last chapter, started out by trying to set a standard with its technology. Only after that effort did not succeed did it shift to an aggressive licensing model to generate its profits.

The IBM personal computer is another example of this tension. The PC triumphed as the architecture that defined the evolution of personal computing since its introduction into the market in 1981. And for the next few years, IBM watched its architecture move its market share past Tandy, past Commodore, and even past Apple to become the dominant share supplier of PCs by 1985.

IBM also had crafted significant protection of key portions of its intellectual property for its PC. It had a strong brand, both for its corporate identity and for its PC product (those readers past a certain age may recall the Charlie Chaplin–esque, “little tramp” advertising theme). It also had patents on many elements of its computer design and carefully fenced off its ROM BIOS code that connected the DOS operating system with the PC hardware. As it happened, companies such as Compaq and Phoenix Software realized that they could legally reverse-engineer this code without violating IBM’s intellectual property. While this was expensive and time consuming, once they succeeded in this effort, dozens of companies (which bought ROM BIOS chips from Compaq and Phoenix Software) could now offer truly compatible computers to IBM.

Thus IBM won the battle of becoming the dominant design. In 2004, however, the company threw in the towel and sold its PC business to Linova, a PC manufacturer based in China. The war *within* the IBM PC industry architecture over the long run had been won by others, such as Dell and HP (which bought Compaq). IBM’s efforts to manage its IP succeeded in establishing the dominant design. Those efforts failed to enable IBM to protect enough of its technology to retain industry leadership two decades later.¹⁴

The iPod must also strike the right balance, now that the device is strongly in its growth phase. If it loses control of its architecture entirely, it could face the fate of the IBM PC. If it keeps too tight a grip on its proprietary approach, other, more open approaches will eventually overtake it in the market (as happened back in the 1980s with the Macintosh versus the PC).

Mature Phase

In the mature phase, the bulk of the market growth has already occurred. Instead of positioning itself for the future, the company must now focus on how to harvest the fruits of the technology today. As product lives shorten in specific markets, maturation encourages companies to begin to entertain other uses of the technology in newly emerging segments within the industry, and even in wholly different uses in other industries.

In the technology life cycle literature, this is the juncture where the basis of technical competition shifts from superior product technology to superior process technology. In part, this is due to the increasing difficulty in differentiating one's products from those of rivals. In part, however, there is now enough widely shared information throughout the industry that vertically integrated organizational structures confer fewer benefits, while more specialized and focused parties within the value chain (with highly optimized processes for those elements of the value chain) become increasingly competitive.

This phase ushers in a variety of innovations in business models, even as the technological differences diminish. Companies now craft many alternative business models, often involving new arrangements with competitors, customers, suppliers, and third parties in other markets. In the PC market, for example, the Dell Direct model emerged to become a dominant force as the industry matured. Taiwanese companies began to manufacture PCs and laptops to the specifications of the big-name computer manufacturers. Other companies started to reverse-engineer the printer cartridges of successful companies like HP.

This phase also ushers in a different objective for managing intellectual property. It is no longer enough for IP management to establish and defend the company's business model within the industry. IP management must now support the application of IP to new segments within the current industry and seek out new applications of the IP in other industries. Polaroid, for example, pushed its instant photography into many commercial applications, such as identity badges and security documents. Kodak for its part pushed its film technology into many parts of medicine, to provide high-quality images for patient diagnosis.

In many cases, those applications will not be pursued by the company owning the IP but instead will be realized by other companies. Overly restrictive IP management strategies will impair the search for these new uses of the company's technology. Proactive and appropriate strategies, however, can help the company identify and exploit these new opportunities. Business models for these new areas of application must take account of the value provided by the new application partner, as well as that of the technology provider.

In later stages of its evolution, RealNetworks has moved increasingly into content and services, now relying on exclusivity in access to content for particular display media (such as PCs and wireless devices such as cell phones). As streaming media has matured, Real has moved its player software (as opposed to its content) toward an open source approach. This increases the number of developers building on Real's technology, even as it loosens Real's ability to control the direction of the RealPlayer software technology.

It is likely that Apple's iPod will have the opportunity to be useful in a variety of applications well beyond the playback of music and video. But Apple will have to be quite agile and adaptable to find the right mix of protection and openness to penetrate these newer application opportunities in a profitable way.

Decline Phase

In the final phase of the technology life cycle, the technology no longer has much, if any, value in the original market. This might be forced by the expiration of legal protection for the IP. More commonly, though, in rapidly advancing industries, the market value of a technology expires long before its legal protection lapses. This typically occurs when there is the introduction of a new, improved technology that replaces the earlier technology. As product lives shorten in many industries, this latter force increasingly comes to the fore.

At first blush, the decline phase might seem like the least interesting phase of the cycle. The key battles have been fought, the winners have emerged, and there is little or no prospect of additional market growth. Meanwhile, newer technologies are entering into the market and obsoleting much, or all, of the value of the current technologies.

In fact, though, alert managers are crafting business models that employ many profitable ways to extract value from IP that might seem to be of little value. GE exited most of its electronics businesses (televisions, radios, stereos, etc.) a long time ago. Yet one can still readily purchase GE brand products of many kinds, thanks to a clever deal GE did with some Asian manufacturers to supply these goods under the GE nameplate. The manufacturers do all the R&D, manufacturing, marketing, sales, and distribution, and pay GE a royalty for the use of its IP (here, the GE trademark). GE takes no business risk, requires no assets to support this use of its IP, and receives a nice royalty for its name. This is a very profitable business model indeed! And it came from GE's choosing to exit a declining set of businesses that it felt were no longer strategically attractive for the company.

IBM is another company that has made some very profitable lemonade by getting out of businesses it felt had become lemons. When IBM exited the PC business, it found that it had some tremendous IP assets that it could leverage against the leaders in the PC business, such as Dell and Linova. In the case of Dell, IBM packaged its IP assets together with a long-term deal to supply various IT services to Dell and announced a relationship that will result in many tens of millions of dollars being paid to IBM. In the case of Linova, IBM also receives more than \$1.7 billion in payment, plus expanded access in China for IBM's IT services businesses. By closing one door (its PC business), IBM has opened another door (its IT services offerings in China). A third exit came in the switches and router business, where IBM was competing against Cisco Systems without much success. When IBM closed its business, it licensed some of its IP to Cisco (again bundled into an announced agreement to sell IT services to Cisco) and received many tens of millions of dollars.

These examples illustrate the value of alert IP management in the declining phase of the IP life cycle model. One factor that adds to the opportunity for value realization in this stage is the fact that the firm is no longer going to be operating in that business. While the firm was in that business, its own assets and operations were at risk of violating another company's IP rights. If the firm elected to assert its IP claims over another firm, that other firm often chose to file a counterclaim against the first firm. That counterclaim alleged that

the first firm was infringing on the second firm's IP. The usual course for such issues is a good deal of posturing by both sides, followed by a settlement whereby each side cross-licenses its IP to the other, perhaps with a payment to make up any perceived difference. This was the situation in figure 4-2.

Once the first firm chooses to leave the business, the situation changes. Now the firm no longer has its own assets exposed to the potential counterclaims of the other party. The apparent weakness (namely, the decision to leave the business) creates a stronger position from which to negotiate with others that might infringe on the firm's IP (which, after all, remains in force even after the firm leaves the business). This is the situation in figure 4-3.¹⁵

Another example of finding IP value in the declining phase of the TLC comes from failed start-up ventures. From 1998 to early 2001, the business world witnessed a boom in start-ups (during the so-called bubble years) that was followed by a corresponding bust. Ventures that had achieved pre-money valuations of \$100 million or more had to sell off all of their assets at auction to recover a few pennies on the dollar for their investors. These assets included the IP held by the venture. While many of these ventures had no assets of any real value, some of them may have had patents on good technologies and ideas that failed due to other reasons. These IP assets are typically auctioned off with the furniture by the auctioneer, such that alert buyers can obtain legally valid patents for as little as a few thousand dollars each. By comparison, the process of developing and filing for a new patent that eventually issues is somewhere between \$15,000 and \$50,000. It is often more expensive to analyze the patents carefully than it is to simply purchase them. Established companies are now rounding out their IP portfolios by shopping at these auctions to acquire additional patents.

Managing Across Phases of the Technology Life Cycle: Windows in China

With the IP life cycle model, the prevailing conception of managing intellectual property with a one-size-fits-all mentality is swept aside. In its place, a more nuanced, dynamic, and strategically focused